

TOPIC 2: COMPONENTS OF THE ENVIRONMENT

1. Biotic and Abiotic Components

Biotic – living things: plants, animals, microorganisms. **Abiotic** – non-living: air, water, soil, sunlight, temperature, minerals.

2. Interdependence

Living things depend on non-living things (e.g. plants need sunlight, water and minerals). Organisms also depend on one another (food, shelter, pollination). This interdependence forms ecosystems.

3. Food Chains and Food Webs

A **food chain** shows a single path of energy flow: producer → primary consumer → secondary consumer → ... A **food web** is a network of interconnected food chains. Producers (plants) make food; consumers eat other organisms; decomposers break down dead matter.

4. Energy Flow in Ecosystems

Energy from the sun is captured by producers and passed along the food chain. At each level some energy is lost as heat, so food chains are limited in length. Decomposers return nutrients to the soil.

5. Summary

The environment has biotic and abiotic parts that interact. Food chains and webs show how energy and nutrients move through ecosystems.

End of Topic 2